

KENOSHA LH, WI, USA

WMO: 997340

Lat: 42.5888N

Lon: 87.8085W

Elev: 196

StdP: 98.99

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.6	(c) -17.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 13.7	(k) -3.0	(l) 12.6	(m) -3.6	(n) 6.2	(o) 280	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.3	(c) 30.8	(d) N/A	(e) 28.7	(f) N/A	(g) 26.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.2	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.8	(b) 12.0	(c) 10.7	(e) -23.1	(f) 33.8	(g) 3.6	(h) 2.3	(i) -25.7	(j) 35.5	(k) -27.9	(l) 36.8	(m) -29.9	(n) 38.1	(o) -32.5	(p) 39.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-5.2	-4.6	1.5	6.9	12.2	17.9	21.9	21.6	17.7	11.5	4.5	-1.1	(6)	
(7)		DBStd	10.52	6.77	6.02	5.27	3.76	4.10	3.89	3.41	2.70	3.69	4.30	5.53	5.62	(7)	
(8)		HDD10.0	1834	470	408	270	108	23	0	0	0	1	35	176	344	(8)	
(9)		HDD18.3	3782	728	641	523	342	196	54	8	6	54	215	414	602	(9)	
(10)		CDD10.0	1402	0	0	6	16	91	238	369	358	231	81	12	1	(10)	
(11)		CDD18.3	309	0	0	1	0	5	42	118	106	35	2	0	0	(11)	
(12)		CDH23.3	1550	0	0	1	4	56	226	698	414	142	9	0	0	(12)	
(13)		CDH26.7	411	0	0	0	1	13	55	218	93	31	1	0	0	(13)	
(14)	Wind	WSAvg	5.2	5.7	5.8	5.3	6.0	5.2	4.5	4.2	4.3	4.9	5.5	5.7	5.6	(14)	
(15)	Precipitation	PrecAvg	878	44	38	55	95	97	104	84	94	81	70	65	49	(15)	
(16)		PrecMax	1056	94	90	111	181	268	217	185	285	176	164	136	103	(16)	
(17)		PrecMin	610	9	4	15	46	16	17	42	36	17	24	15	16	(17)	
(18)		PrecStd	116	23	24	30	38	53	50	33	48	39	39	35	24	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	12.0	20.8	22.7	28.6	32.0	35.0	32.1	30.0	24.7	18.7	13.9	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	8.9	7.6	15.5	18.7	24.8	28.7	31.4	29.4	26.9	21.5	16.1	9.7	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	5.8	4.8	11.3	15.1	21.8	26.3	29.2	27.0	24.7	19.0	13.9	7.6	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	3.8	2.6	7.8	12.0	18.9	24.2	27.1	25.4	22.9	17.5	11.7	5.8	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR		7.3	6.9	6.0	6.2	7.3	6.9	6.3	5.7	6.4	6.6	6.5	6.4	(35)	
(36)		5% DB	MCD BR	9.4	9.4	11.3	12.0	13.1	10.4	9.8	8.4	9.0	9.1	9.0	7.5	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCD BR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.293	0.310	0.343	0.394	0.419	0.437	0.435	0.428	0.399	0.351	0.313	0.292	(40)	
(41)		taud		2.450	2.401	2.397	2.287	2.268	2.267	2.272	2.291	2.345	2.476	2.524	2.491	(41)	
(42)		Ebn at Noon		853	892	901	874	857	840	837	831	829	835	825	823	(42)	
(43)		Edh at Noon		75	92	105	127	133	133	131	125	110	86	70	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.46	2.34	3.54	4.59	5.28	6.04	6.14	5.30	4.30	2.78	1.73	1.20	(44)	
(45)		RadStd		0.19	0.28	0.30	0.38	0.41	0.40	0.28	0.33	0.36	0.32	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page